2-Adic Local Constancy of Transfer Matrices for Generalized Collatz Maps

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Abstract

We study the parametric family of Syracuse-type maps $S(n) = (xn + y)/2^{v_2(xn+y)}$ on odd integers, where x and y are odd parameters. At resolution k, the map acts on odd residues modulo 2^k via a transfer matrix $P_k(x, y)$. We prove that the map $x \mapsto P_k(x, y)$ is locally constant in the 2-adic topology, with tight modulus $M = k + V$ where V is the maximum 2-adic valuation across columns. This result shows that polynomial continuation of spectral data in the multiplier x is impossible — the natural domain is the 2-adic integers, not the complex plane. For the classical Collatz case $x = 3$, $y = 1$, we give explicit bounds $M \approx 2k$, with worst-case residues $j_m = (4^m - 1)/3$. An analogous result holds for the shift parameter y. Computation confirms $M \approx 2k$ for all odd $x \in \{3, 5, \ldots, 21\}$.

1 Introduction

The Collatz conjecture asserts that iterating $n \mapsto n/2$ (if n is even) or $n \mapsto (3n + 1)/2$ (if n is odd) eventually reaches 1 from any positive starting value. We embed this problem in a parametric family: for odd positive integers x and y, define the Syracuse-type map on odd integers:

$$S(n) = \frac{xn + y}{2^{v_2(xn+y)}},$$

where $v_2(m)$ denotes the 2-adic valuation of m. The classical Collatz map corresponds to $x = 3$, $y = 1$.

At resolution $k \geq 2$, the map S acts on the $N = 2^{k-1}$ odd residue classes modulo 2^k, encoded in a transfer matrix $P_k(x, y)$ whose spectral radius $\rho_k(x, y)$ measures the worst-case contraction rate over all modular cycles. A natural question is: how does P_k depend on the multiplier x? This is motivated by the discovery that the exceptional set E — levels where additional modular cycles appear — is infinite with positive density (McKenna, 2026). One might hope for polynomial dependence in x, enabling analytic continuation techniques. Our main result shows the dependence is fundamentally different: P_k is a step function that jumps at 2-adic boundaries.

This connects to the framework of Siegel (2025), who treats Collatz-type maps as iterated function systems on 2-adic integers. Our result places the transfer matrix naturally within the class of locally constant functions on $\mathbb{Z}_2$ that his framework is designed to handle.

Prior work. Matthews and Watts (1985) studied generalized Collatz mappings. Kontorovich and Lagarias (2009) compared $3x + 1$ and $5x + 1$ using growth-rate criteria. Matthews (2010) constructs row-stochastic transition matrices on residues. Our transfer matrix differs in that entries are weighted by 2^{-v}, making the spectral radius directly the convergence indicator.

Notation. Throughout, x denotes an odd positive integer (the multiplier), y an odd integer (the shift), $k \geq 2$ an integer (the resolution), $v_2(m)$ the 2-adic valuation of m, and $\rho_k(x, y)$ the spectral radius of $P_k(x, y)$.

Outline. Section 2 constructs the transfer matrix and recalls standard spectral properties. Section 3 states and proves the main theorem. Section 4 derives corollaries including non-polynomiality of Fredholm coefficients and an analogous result for the shift parameter y. Section 5 gives explicit bounds for the Collatz case. Section 6 discusses consequences and open questions.

2 The Transfer Matrix

2.1 Construction

Fix $k \geq 2$, x odd positive, and y odd. Let $N = 2^{k-1}$, and let $R = \{1, 3, 5, \ldots, 2^k - 1\}$ be the odd residues modulo 2^k, indexed as $r_i = 2i + 1$ for $i = 0, \ldots, N - 1$.

Definition 1. *The **transfer matrix** $P_k(x, y)$ is the $N \times N$ matrix defined as follows. Columns and rows are indexed by the odd residues $j \in R = \{1, 3, \ldots, 2^k - 1\}$, with the column-to-index map* $\mathrm{idx}(j) = (j - 1)/2$. *For each column $j \in R$, compute:*

$$\mathrm{val}_j = x \cdot j + y, \tag{1}$$

$$v_j = v_2(\mathrm{val}_j), \tag{2}$$

$$t_j = (\mathrm{val}_j / 2^{v_j}) \bmod 2^k. \tag{3}$$

Set $P[t_j, j] = 2^{-v_j}$ (using odd residues as row and column labels). All other entries in column j are zero.

The matrix is well-defined: since x, r_j, and y are all odd, $\mathrm{val}_j = x \cdot r_j + y$ is even (odd $\times$ odd $+$ odd $=$ even), so $v_j \geq 1$, and t_j is odd by maximality of v_j.

Each column has exactly one nonzero entry, so P_k encodes a function on R: the Syracuse map S reduced modulo 2^k, with each transition weighted by its contraction factor 2^{-v_j}.

2.2 Spectral Radius

Since each column has exactly one nonzero entry, the functional graph of P_k decomposes into cycles and trees rooted at those cycles. The eigenvalues of P_k are determined by the cycles: each cycle of length L with valuations $v_1, \ldots, v_L$ contributes an eigenvalue

$$\lambda = \left(\prod_{i=1}^{L} 2^{-v_i} \right)^{1/L} = 2^{-\bar{v}},$$

where $\bar{v} = \frac{1}{L} \sum_{i=1}^{L} v_i$ is the mean valuation around the cycle. The spectral radius is therefore

$$\rho_k(x, y) = \max_{\text{cycles}} 2^{-\bar{v}}.$$

Standard bounds. Since $v_j \geq 1$ for all columns (as x, r_j, y are all odd), $\rho_k(x, y) \leq 1/2$ for all odd x, y (cf. Lagarias, 1985, Proposition 1, for the $x = 3$ case; the same argument applies generally). The 2-adic valuation over odd residues follows the universal geometric distribution $P(v = j) = 1/2^j$ for $j = 1, \ldots, k - 1$ (folklore; see Tao, 2022; Matthews, 2010). The Fredholm zeros $z_0 = 1/\lambda_i$ all satisfy $|z_0| \geq 2$, trivially from the spectral bound.

3 2-Adic Local Constancy

Theorem 1 (2-Adic Local Constancy). *For fixed $k \geq 2$ and y odd, the map $x \mapsto P_k(x, y)$ is locally constant in the 2-adic topology on the odd positive integers. Explicitly: for each odd positive integer x_0,*

$$P_k(x, y) = P_k(x_0, y) \quad \text{for all odd positive } x \text{ with } x \equiv x_0 \pmod{2^M},$$

where

$$M(k, x_0, y) = k + V(k, x_0, y), \qquad V(k, x_0, y) = \max\{v_2(x_0 j + y) : j \text{ odd},\ 1 \leq j < 2^k\}.$$

This bound is tight: no smaller M suffices.

Proof. We must show that if $x \equiv x_0 \pmod{2^M}$ with $M = k + V$, then $P_k(x, y) = P_k(x_0, y)$. Since the matrix is determined column by column, it suffices to show that for each odd $j \in R$, the pair (v_j, t_j) is the same for x and x_0.

Fix an odd residue $j \in R$. Let $\mathrm{val} = x_0 j + y$ and $\mathrm{val}' = xj + y = \mathrm{val} + (x - x_0)j$. Write $v = v_2(\mathrm{val})$, so $\mathrm{val} = 2^v q$ with q odd. The perturbation is $\delta = (x - x_0)j$.

Since $x \equiv x_0 \pmod{2^M}$ and j is odd:

$$v_2(\delta) = v_2(x - x_0) + v_2(j) = v_2(x - x_0) \geq M.$$

Step 1 (Valuation preservation). We have $\mathrm{val}' = \mathrm{val} + \delta = 2^v q + \delta$. Since $v_2(\delta) \geq M > v$ (because $M = k + V \geq k + v$ and $k \geq 2$), the ultrametric identity gives

$$v_2(\mathrm{val}') = v_2(\mathrm{val} + \delta) = \min(v_2(\mathrm{val}), v_2(\delta)) = v.$$

The identity $v_2(a + b) = \min(v_2(a), v_2(b))$ holds whenever $v_2(a) \neq v_2(b)$. Therefore v_j is the same for x_0 and x, and the weight 2^{-v_j} is unchanged.

Step 2 (Target preservation). The target is $t = (\mathrm{val}/2^v) \bmod 2^k$. We have:

$$\mathrm{val}'/2^v = \mathrm{val}/2^v + \delta/2^v = q + \delta/2^v.$$

Since $v_2(\delta) \geq M = k + V \geq k + v$, we have $v_2(\delta/2^v) \geq k$, so

$$\delta/2^v \equiv 0 \pmod{2^k},$$

giving $\mathrm{val}'/2^v \equiv \mathrm{val}/2^v \pmod{2^k}$.

Since both v_j and t_j are preserved for every column j, the entire matrix is unchanged. □

Minimality. Consider the column j^* achieving $V = v_2(x_0 j^* + y)$, and set $M' = k + V - 1$. Taking $x = x_0 + 2^{M'}$, the perturbation $\delta = 2^{M'} j^*$ has $v_2(\delta) = M'$. Since $M' = k + V - 1 > V = v_2(\mathrm{val}_{j^*})$, the valuation is preserved. But $\delta/2^V$ has $v_2 = k - 1$, so $\delta/2^V \not\equiv 0 \pmod{2^k}$: the target t_{j^*} changes modulo 2^k. Therefore $P_k(x, y) \neq P_k(x_0, y)$. □

4 Corollaries

The following concern the Fredholm determinant $F_k(z; x, y) = \det(I - zP_k(x, y))$.

Corollary 1. *The Fredholm determinant* $\det(I - zP_k(x,y))$*, viewed as a function of* x *for fixed* k*,* y*,* z*, is locally constant on the odd 2-adic integers.*

Corollary 2. *The spectral radius* $\rho_k(x,y)$ *is locally constant in* x *(2-adic topology).*

Corollary 3 (Finiteness)**.** *For fixed* k *and* y*, the spectral radius* $\rho_k(x,y)$ *takes at most* 2^{M-1} *distinct values as* x *ranges over odd positive integers, where* $M = M(k, x_0, y)$ *is the modulus of the constancy class containing* x_0*. For* $x = 3$*,* $y = 1$*, we have* $M \leq 2k+2$ *(Section 5); computationally,* $M \approx 2k$ *holds for all odd* $x \leq 21$ *(Table 2).*

Corollary 4 (Non-polynomiality)**.** *The Fredholm coefficients* $c_j(x)$ *are not polynomial functions of* x *for* $j \geq 1$*.*

Proof. At $k = 3$, $c_1(x) = -\operatorname{tr}(P_3(x,1))$. For $x = 19$, direct computation gives $c_1(19) = 0$ (no odd residue modulo 8 maps to itself under $S_{19,1}$). Since there are finitely many constancy classes for $P_3(x,1)$ and each class is attained by infinitely many odd x (by local constancy), infinitely many odd x satisfy $c_1(x) = 0$. But $c_1(3) = -0.25 \neq 0$. A polynomial with infinitely many roots is identically zero, contradicting $c_1(3) \neq 0$. □

Remark. *This result identifies the correct topology for extending spectral data in* x*: the natural domain is the 2-adic integers* $\mathbb{Z}_2$*, not the complex plane. The polynomial interpolation approach (fitting Fredholm coefficients at integer* x *and evaluating at complex* x*) is the wrong tool. This connects to Siegel's framework (2025), which treats Collatz-type maps as iterated function systems on* $\mathbb{Z}_2$*.*

The proof of Theorem 1 applies verbatim to the shift parameter y, giving a companion result:

Proposition 4 (y-Local Constancy)**.** *For fixed* $k \geq 2$ *and* x *odd, the map* $y \mapsto P_k(x,y)$ *is locally constant in the 2-adic topology on the odd integers. Explicitly, for each odd* y_0*,*

$$P_k(x,y) = P_k(x,y_0) \quad \textit{for all odd } y \textit{ with } y \equiv y_0 \pmod{2^{M_y}},$$

where $M_y = k + V(k,x,y_0)$*, with* V *as in Theorem 1. This bound is tight.*

Proof. Fix column $j \in R$ (odd). The value is $\mathrm{val} = xj + y_0$, and the perturbation from replacing y_0 by y is $\delta = y - y_0$, independent of j. Since $v_2(\delta) \geq M_y = k + V \geq k + v_j$, the ultrametric identity gives $v_2(\mathrm{val} + \delta) = v_j$, and $\delta/2^{v_j}$ has $v_2 \geq k$, so $(\mathrm{val}+\delta)/2^{v_j} \equiv \mathrm{val}/2^{v_j} \pmod{2^k}$. Both the weight 2^{-v_j} and the target t_j are preserved for every column. Minimality: take $y = y_0 + 2^{M_y - 1}$; at the column j^* achieving V, $\delta/2^V$ has $v_2 = k-1$, so t_{j^*} changes modulo 2^k. □

Corollary 5 (Joint Local Constancy)**.** *The map* $(x,y) \mapsto P_k(x,y)$ *is locally constant on* $\mathbb{Z}_2^\times \times \mathbb{Z}_2^\times$ *(the 2-adic units, i.e., odd 2-adic integers) with the product 2-adic topology.*

Proof. Perturbations $\delta_x j$ and δ_y are controlled independently by Theorem 1 and Proposition 4; their sum satisfies the same ultrametric bound. □

5 Explicit Bounds for $x = 3$, $y = 1$

The maximum $V(k,3,1) = \max\{v_2(3j+1) : j \text{ odd}, 1 \leq j < 2^k\}$ is achieved when $3j+1$ is a power of 2. Setting $3j + 1 = 2^s$ gives $j = (2^s - 1)/3$, which is an integer when s is even. Writing $s = 2m$ gives $j_m = (4^m - 1)/3$, which is always odd (since $j_m = \sum_{i=0}^{m-1} 4^i \equiv 1 \pmod 2$).

The sequence of worst-case residues is $j_1 = 1, j_2 = 5, j_3 = 21, j_4 = 85, j_5 = 341, \ldots$, giving $v_2(3j_m + 1) = 2m$. The largest $j_m < 2^k$ satisfies $m < (k + \log_2 3)/2$, so $V \approx k$ and $M \approx 2k$.

Table 1: Explicit modulus bounds for the Collatz case $x = 3$, $y = 1$. Any $x \equiv 3 \pmod{2^M}$ (odd) gives $P_k(x, 1) = P_k(3, 1)$.

k	$V = \max v_2$	$M = k + V$	Worst j
3	4	7	5
5	6	11	21
7	8	15	85
9	10	19	341

The worst-case residues $j_m = (4^m - 1)/3$ exhibit a structural pattern: each is the m-fold iterate of the map $j \mapsto 4j + 1$ starting from $j_1 = 1$. These are precisely the residues where the Syracuse map produces maximum 2-adic cancellation.

Remark. *The growth $M \approx 2k$ means that determining the transfer matrix at resolution k requires knowing approximately $2k$ bits of the multiplier. This is not merely an artifact of the proof — the minimality argument shows that $M - 1$ bits genuinely do not suffice.*

6 Discussion

6.1 The Natural Domain Is $\mathbb{Z}_2$, Not $\mathbb{C}$

Theorem 1 shows that $P_k(x, y)$ is a *step function* of x in the 2-adic topology: constant on each coset of $2^M\mathbb{Z}$ within the odd integers, with step boundaries at 2-adic neighborhoods. This has a negative and a positive consequence.

Negative. Any attempt to extend the spectral data $\rho_k(x, y)$, or the Fredholm coefficients $c_j(x)$, as analytic functions of $x \in \mathbb{C}$ is impossible. Corollary 4 gives non-polynomiality directly. More generally, a 2-adically locally constant non-constant function cannot be the restriction of any holomorphic function on a connected domain in $\mathbb{C}$: 2-adic constancy classes are dense in every real interval (since $\{x : x \equiv a \pmod{2^m}\}$ is dense in $\mathbb{R}$ for any a), so a 2-adically locally constant non-constant function is discontinuous everywhere in the Archimedean topology, and hence cannot be the restriction of any continuous (let alone holomorphic) function on a real interval.

Positive. The 2-adic topology is the *correct* topology for studying parametric variation in x. Any result established for x_0 holds automatically for all $x \equiv x_0 \pmod{2^M}$. Properties proved for $P_k(3, 1)$ extend to all $x \equiv 3 \pmod{2^M}$. This connects naturally to Siegel (2025), who works exclusively in the 2-adic category; Theorem 1 provides an independent elementary proof that the 2-adic framework is the correct setting for the multiplier variable.

6.2 The Phase Transition at $x = 4$

The expected log-growth per Syracuse step is $\log_2 x - \mathbb{E}[v] = \log_2 x - 2$ (from $\mathbb{E}[v] = \sum_{j \geq 1} j \cdot 2^{-j} = 2$), changing sign at $x = 4$: orbits are expected to shrink for odd $x \leq 3$ and grow for odd $x \geq 5$. The 2-adic distance from $x = 3$ to $x = 5$ is $|3 - 5|_2 = 2^{-1}$, the maximum between consecutive odd integers, and $P_k(3, y) \neq P_k(5, y)$ for all $k \geq 2$. The phase transition is thus a discrete jump between two locally constant regimes, with no interpolation through the (undefined) even value $x = 4$.

The contraction-weighted matrices studied here satisfy $\rho_k \leq 1/2$ universally (Proposition 1 of McKenna, 2026), regardless of growth-weighted behavior. The spectral-theoretic realization of the phase transition must be understood 2-adically: the relevant parameter space is not the real line but $\mathbb{Z}_2^\times$, where $x = 3$ and $x = 5$ are adjacent neighbors rather than straddling a transition point.

6.3 Connection to Ghost Cycles

In McKenna (2026), *ghost cycles* arise at exceptional levels $k \in E$ and are classified by the denominator $D = 2^V - 3^L$. For general odd x, the denominator becomes $D(x) = 2^V - x^L$. Theorem 1 implies that for fixed k and y, the entire cycle structure of $P_k(x, y)$ — number of cycles, lengths, valuation patterns, denominators — is constant on each 2-adic neighborhood. In particular:

(a) Ghost types are locally constant in x. If a ghost with parameters $(L, V, (v_1, \ldots, v_L))$ exists at (x_0, k), it exists for all odd $x \equiv x_0 \pmod{2^M}$.

(b) Spectral contributions are combinatorially determined. Each cycle contributes $2^{-V/L}$ to the spectral radius; this depends on (L, V) alone, not on x.

(c) Persistence periods are locally constant. The period $p(x) = \mathrm{ord}_2(|D(x)|)$ at which a ghost type reappears is itself locally constant in x. Write $D(x) - D(x_0) = x_0^L - x^L = (x_0 - x)(x_0^{L-1} + \cdots + x^{L-1})$. For odd L, the second factor is a sum of L terms each $\equiv x_0^{L-1} \pmod 2$ (hence odd), so $v_2(D(x) - D(x_0)) = v_2(x - x_0)$. Taking $v_2(x - x_0) > v_2(D(x_0))$ ensures $v_2(D(x)) = v_2(D(x_0))$ by the ultrametric identity, hence $p(x) = p(x_0)$. For even L, the second factor has an additional factor of 2, but the argument proceeds similarly with $M' = v_2(D(x_0)) + 1$.

The ghost cycle structure is therefore a 2-adic invariant of x: 2-adically close multipliers share ghost types, exceptional set structure, and persistence periods.

6.4 Implications for Spectral Theory

Since $\rho_k(x, y)$ is locally constant (Corollary 2), any proof that $\rho_k < 1/2$ at x_0 extends immediately to a 2-adic neighborhood of x_0. For the infinite-dimensional transfer operator $\mathcal{L}$ (defined in McKenna, 2026) on $C(\mathbb{Z}_2^\times)$, the spectral radius $\rho(\mathcal{L})(x) = \sup_k \rho_k(x, y)$ is lower semicontinuous in x (supremum of locally constant functions is lower semicontinuous). It need not be locally constant itself: the constancy balls shrink as k grows (since $M \approx 2k$), so the limit could vary on arbitrarily fine 2-adic scales.

Local constancy also reduces the parameter space constructively. To determine ρ_k for all odd x, it suffices to compute P_k at $O(4^k)$ representative values (Corollary 3). For $y = 1$ and $k = 3, 4, 5, 6$, direct computation over odd $x \in [1, 2^{2k+2})$ shows that the number of distinct transfer matrices is exactly 2^{2k} in all four cases, attaining the Corollary 3 bound up to a factor of 2.

6.5 Open Questions

1. Growth rate of M for general (x, y). For $x = 3$, $y = 1$, Section 5 proves $M \approx 2k$. The efficient formula $j^* = -yx^{-1} \bmod 2^k$ (the unique maximizer of $v_2(xj+y)$ over odd $j < 2^k$) enables large-scale computation: for $x \in \{3, 5, 7, \ldots, 21\}$ and $k = 3, \ldots, 30$, fitting $V(k, x, 1) \approx c(x, 1) \cdot k$ gives the values in Table 2. All slopes satisfy $c(x, 1) \approx 1.00$ (range 0.984–1.014, $R^2 > 0.96$), confirming that $M \approx 2k$ is not special to $x = 3$ but holds across odd multipliers. The heuristic: $j^* = -yx^{-1} \bmod 2^k$ is a 2-adic unit, so generically $v_2(xj^* + y) \approx k$, giving $V \approx k$ and $M \approx 2k$. Proving this rigorously for all odd (x, y) is open.

Table 2: Fitted growth rate $c(x, 1)$ from $V(k, x, 1) \approx c \cdot k$ for $k = 5, \ldots, 30$ and $y = 1$. All slopes are near 1, confirming $M \approx 2k$ universally.

x	$c(x,1)$	$M \approx$	R^2
3	0.9956	$1.996k$	0.996
5	1.0027	$2.003k$	0.988
7	0.9969	$1.997k$	0.989
9	1.0106	$2.011k$	0.979
11	0.9949	$1.995k$	0.982
13	0.9843	$1.984k$	0.983
15	0.9956	$1.996k$	0.978
17	1.0140	$2.014k$	0.964
19	0.9959	$1.996k$	0.981
21	1.0051	$2.005k$	0.963

2. Density of constancy classes. Corollary 3 bounds the number of distinct transfer matrices at resolution k by $O(4^k)$. The exact count as $k \to \infty$, and whether any hidden symmetry reduces it substantially below this bound, is open.

3. Local constancy of $\rho(\mathcal{L})$. Since $M(k) \to \infty$, the infinite-dimensional spectral radius is only lower semicontinuous. Whether it is locally constant (or varies on all 2-adic scales) is open, and closely related to whether $\mathcal{L}$ is compact on $C(\mathbb{Z}_2^\times)$ (see McKenna, 2026, Section 7).

4. Limiting Fredholm determinant. Each $\det(I - zP_k(x, y))$ is locally constant in x (Corollary 1). If $F(z; x, y) = \lim_{k\to\infty} \det(I - zP_k(x, y))$ exists (related to compactness of $\mathcal{L}$), then local constancy of F would mean the full eigenvalue spectrum is a 2-adic invariant of x.

5. Extension to 2-adic multipliers. Since odd positive integers are dense in $\mathbb{Z}_2^\times$, Theorem 1 and Proposition 4 extend by continuity: for any $\xi \in \mathbb{Z}_2^\times$, define $P_k(\xi, y) = P_k(x_0, y)$ for any odd integer $x_0 \equiv \xi \pmod{2^M}$; local constancy ensures consistency. This defines the transfer matrix for non-integer 2-adic multipliers, connecting to the shifted-coordinate representation $m = n + 1/3$ that makes the Syracuse map multiplicative (cf. Kontorovich and Lagarias, 2009).

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